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One more result showing that "every matrix is almost diagonalizable".

<u>Theorem 2</u> Let $A \in M_n$. Then for all $\varepsilon > 0$ there exists invertible $S_\varepsilon \in M_n$ such that

$$S_\varepsilon^{-1} A S_\varepsilon = T_\varepsilon = [t_{ij}(\varepsilon)]$$

is upper triangular and for all $i<j$,

$$|t_{ij}(\varepsilon)| < \varepsilon$$

(i.e. entries in upper triangular part are arbitrarily small)

<u>Proof</u> Start with Schur's Theorem:

$$U^* A U = T, \text{ where}$$

U is unitary & $T = [t_{ij}]$ is upper triangular.

Let

$$t = \max_{i<j} \{|t_{ij}|\}$$

<u>Notation</u> For every $\alpha \in \mathbb{C} \setminus \{0\}$, denote

$$D_\alpha = \operatorname{diag}(1, \alpha, \alpha^2, \dots, \alpha^{n-1})$$

$$= \begin{bmatrix} 1 & & & 0 \\ & \alpha & & \\ & & \alpha^2 & \\ 0 & & & \ddots \; \alpha^{n-1} \end{bmatrix} \in M_n.$$

Consider now $0 < \varepsilon < 1$.

If $\begin{cases} t \le 1, & \text{let } S_\varepsilon = U D_\varepsilon \\ t > 1, & \text{let } S_\varepsilon = U D_{1/t} D_\varepsilon \end{cases}$

In either case, one (i.e you!) can verify that

$$S_\varepsilon^{-1} A S \quad \text{is as desired } (T(\varepsilon))$$

$$\left(S_\varepsilon^{-1} = D_\varepsilon^{-1} U^* \text{ or } S_\varepsilon^{-1} = D_\varepsilon^{-1} D_{1/t}^{-1} U^*, \right.$$

$$\left. D_\varepsilon^{-1} = D_{1/\varepsilon}, \; D_{1/t}^{-1} = D_t \right)$$

(e.g. if $t \le 1$, entries of $S_\varepsilon^{-1} A S_\varepsilon$ are

$$\left(\left| \varepsilon^{-i} t_{ij} \varepsilon^{j} \right| < \varepsilon \right) \blacksquare$$

<u>Summarize</u>

Theorem 1

Every A is arbitrarily close in Frobenius norm to a matrix with distinct eigenvalues & thus diagonalizable

Theorem 2

Every A is similar to a triangular matrix whose upper triangular entries are arbitrarily close to 0

One last implication of Schur's Theorem (5)

(σ: spectrum)

Theorem (amazing!)

Let $A, B \in M_n$, $\sigma(A) = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$, $\sigma(B) = \{b_1, b_2, \ldots, b_n\}$. Assume $\boxed{AB = BA}$.

Then there exist a permutation

$$\{i_1, i_2, \ldots, i_n\}$$

of $\{1, 2, \ldots, n\}$ such that

$$\sigma(A+B) = \{\alpha_1 + b_{i_1}, \alpha_2 + b_{i_2}, \ldots, \alpha_n + b_{i_n}\}$$

(i.e. the eigenvalues of the sum are sums of the eigenvalues of the summands paired in some fixed way)

In particular,

$$\sigma(A+B) \subseteq \underbrace{\sigma(A) + \sigma(B)}_{\text{all possible sums of pairs.}}$$

Comment Powerful, because eigenvalues of sums are intractable functions.

Proof

Recall that we "proved" that commuting matrices are simultaneously triangularizable via Schur's Theorem, i.e.

$$AB = BA \implies \exists \text{ unitary } U \text{ such that}$$

$$U^* A U = T = \text{triangular}$$

$$U^* B U = R = \text{———} \text{ " } \text{———}.$$

So $T = \begin{bmatrix} \alpha_1 & & & * \\ & \alpha_2 & & \\ & & \ddots & \\ 0 & & & \alpha_n \end{bmatrix}$, $R = \begin{bmatrix} \beta_{i_1} & & & * \\ & \beta_{i_2} & & \\ & & \ddots & \\ 0 & & & \beta_{i_n} \end{bmatrix}$

for some permutation.

Then

$$U^*(A+B)U = T + R = \begin{bmatrix} \alpha_1 + \beta_{i_1} & & & * \\ & \alpha_2 + \beta_{i_2} & & \\ & & \ddots & \\ 0 & & & \alpha_n + \beta_{i_n} \end{bmatrix}$$

i.e. $\sigma(A+B) = \{\alpha_1 + \beta_{i_1}, \alpha_2 + \beta_{i_2}, \dots, \alpha_n + \beta_{i_n}\}$. ∎

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Review "power method" for finding eigenvalues

Historically the first method for approximating eigenvalues

Actually, in many respects, the only method!

(All state of the art methods, QR-algorithm, Francis algorithm are essentially iterative power computations)

Set-up $A \in M_n$, $\sigma(A) = \{\lambda_1, \lambda_2, \dots, \lambda_n\}$

Assume $|\lambda_1| > |\lambda_2| > |\lambda_3| > \dots > |\lambda_n|$

(i.e. eigenvalues are distinct)

Theorem! ← As a consequence, there exist lin. independent eigenvectors

$\{x_1, x_2, \dots, x_n\}$

$Ax_j = \lambda_j x_j$, $j = 1, 2, \dots, n$

Let's "find" the eigenpair (λ_1, x_1).

Choose "arbitrary" initial vector $x^{(0)} \in \mathbb{C}^n$
Then there exist $a_1, a_2, \ldots, a_n \in \mathbb{C}$ such that

$$x^{(0)} = a_1 x_1 + a_2 x_2 + \cdots + a_n x_n$$

$$x^{(1)} = A x^{(0)} = a_1 A x_1 + a_2 A x_2 + \cdots + a_n A x_n$$
$$= a_1 \lambda_1 x_1 + a_2 \lambda_2 x_1 + \cdots + a_n \lambda_n x_n$$

In general

$$x^{(k)} = A x^{(k-1)} = A^k x^{(0)} = a_1 \lambda_1^k x_1 + a_2 \lambda_2^k x_2 + \cdots + a_n \lambda_n^k x_n$$

$$k = 1, 2, \ldots$$

Consider

$$\frac{x^{(k)}}{\lambda_1^k} = a_1 x_1 + a_2 \left(\frac{\lambda_2}{\lambda_1}\right)^k x_2 + \cdots + a_n \left(\frac{\lambda_n}{\lambda_1}\right)^k x_n$$

$$\left|\frac{\lambda_j}{\lambda_1}\right| < 1, \quad j = 2, \ldots, n$$

and so $\left(\frac{\lambda_j}{\lambda_1}\right)^k \longrightarrow 0$ as $k \to \infty$

i.e.

$$\frac{x^{(k)}}{\lambda_1^k} \xrightarrow{k \to \infty} a_1 x_1 \in \operatorname{span}\{x_1\}$$

(since $\operatorname{span}\{x_1\}$ is a subspace)

i.e.

$$x^{(k)} \xrightarrow{k \to \infty} \operatorname{span}\{x_1\}.$$

Thus $x^{(k)}$ resembles an eigenvector of λ_1 (multiple of x_1) as $k \to \infty$.

To remedy possible large growth of $x^{(k)}$, we work with the sequence

$$y^{(k)} = \frac{x^{(k)}}{\|x^{(k)}\|_2}, \quad k = 0, 1, \ldots$$

$$y^{(k)} \longrightarrow w = \alpha x_1, \quad \|y^{(k)}\| = \|w\| \cong 1$$

for large k.

Having w, we can estimate λ_1 !!

$$Aw = \lambda_1 w$$

$$\frac{(Aw)_j}{w_j} \cong \lambda_1 \quad \text{for any } w_j \neq 0$$

We achieved an approximation of (λ_1, x_1)

How well it works (fast), depends on

$$\left|\frac{\lambda_j}{\lambda_1}\right|, \quad j = 2, 3, \ldots, n$$

Caution

Choice of $x^{(0)}$ should be of a vector that has a component in the direction of x_1 (i.e. $a_1 \neq 0$). Otherwise

$$x^{(k)} \in \operatorname{span}\{x_2, \dots, x_n\} \text{ for all } k.$$

In practice, this is <u>not</u> a problem. Round-off errors in producing the iterates will introduce (a rapidly growing) component in the direction of x_1

How about the other eigenvalues?

Deflation!

Take $x_1 \rightsquigarrow$ complete to a basis for $\mathbb{C}^n$

$$\{x_1, y_2, y_3, \dots, y_n\}$$

Form. $S = [x_1 | y_2 | \cdots | y_n] \in M_n$

Then

$$S^{-1}AS = \left[\begin{array}{c|c} \lambda_1 & 0 \\ \hline 0 & \\ \vdots & A_1 \\ 0 & \end{array}\right] \quad A_1 \in M_{n-1}$$

since distinct.

where $\sigma(A) = \lambda_1 \cup \sigma(A_1)$

& eigenvectors of A corresponding to $\lambda_2, \dots, \lambda_n$

$(\sigma(A_1) = \{\lambda_2, \lambda_3, \dots \lambda_n\})$

easy to prove $\leftarrow$ are of the form

$$S^{-1}\begin{bmatrix} 0 \\ z_j \end{bmatrix}, \quad z \in \mathbb{C}^{n-1}, j = 2, \dots, n$$

where $A_1 z_j = \lambda_j z_j, \; j = 2, \dots, n.$

We can now apply power method to A_1 to estimate λ_2 & z_2, which will give us an estimation of (λ_2, y_2).

Repeat $\longrightarrow$ "deflation"

How about when the eigenvalues are <u>not</u> distinct/ having a basis of eigenvectors!

Exploit the fact $A \sim$ matrix with distinct eigenvalues

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§ 2.5 Normal matrices

Definition $A \in M_n$ is called normal if $AA^* = A^*A$.

(i.e. normal matrices commute with their conjugate transposes)

Remarks / examples

(1) A is normal iff U^*AU is normal for any unitary U

$$(U^*AU)(U^*AU)^* = U^*A\cancel{UU^*}A^*U$$
$$= U^*AA^*U$$
$$\overset{AA^*=A^*A}{=} U^*A^*AU$$
$$= U^*A^*UU^*AU$$
$$= (U^*AU)^*(U^*AU)$$

Converse similar.

(2) Unitary matrices are normal!

$$U^* = U^{-1}$$

so $U^*U = U^{-1}U = I = UU^{-1} = UU^*$

(3) Hermitian ($A = A^*$) are normal

(4) Diagonal matrices are normal

$$\underbrace{\begin{bmatrix} d_1 & & 0 \\ & \ddots & \\ 0 & & d_n \end{bmatrix}^*}_{D^*} \underbrace{\begin{bmatrix} d_1 & & 0 \\ & \ddots & \\ 0 & & d_n \end{bmatrix}}_{D}$$

$$= \begin{bmatrix} |d_1|^2 & & 0 \\ & \ddots & \\ 0 & & |d_n|^2 \end{bmatrix}$$

$$= DD^*$$

(5) $A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ is normal

$$A^*A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = AA^*$$

and A is <u>not</u> hermitian, not unitary, not diagonal – so there's <u>more</u> to normality.

<u>Theorem</u> (main)

Let $A = [a_{ij}] \in M_n$ and suppose

$$\sigma(A) = \{\lambda_1, \lambda_2, \dots, \lambda_n\}.$$

The following are equivalent:

(a) A is normal

(b) A is unitarily diagonalizable, i.e.

$\exists$ unitary $U \in M_n$ such that

$$U^* A U = D \qquad (U^* = U^{-1})$$

and D is diagonal matrix

(c) $$\underbrace{\sum_{i,j=1}^{n} |a_{ij}|^2}_{\text{Frobenius norm}} = \sum_{j=1}^{n} |\lambda_j|^2$$

(d) There exists an orthonormal set of n eigenvectors of A.

Proof

(a) $\Rightarrow$ (b) By Schur's Theorem, there exists unitary U such that

$$U^* A U = T = [t_{ij}]$$

is triangular. By our observation above, we have T is normal: $T^*T = TT^*$.

In particular

$$(T^*T)_{11} = (TT^*)_{11}$$

$$\Rightarrow \quad t_{11}\bar{t}_{11} = t_{11}\bar{t}_{11} + \sum_{j=2}^{n} t_{1j}\bar{t}_{1j}$$

$$= t_{11}\bar{t}_{11} + \sum_{j=2}^{n} |t_{1j}|^2$$

$$\Rightarrow \quad \sum_{j=2}^{n} |t_{1j}|^2 = 0$$

$$\Rightarrow \quad t_{1j} = 0, \quad j = 2, \dots n$$

i.e.

$$T = \begin{bmatrix} t_{11} & 0 & \cdots & 0 \\ 0 & t_{22} & & * \\ \vdots & & \ddots & \\ \vdots & 0 & & t_{nn} \end{bmatrix}$$

Forcing $(T^*T)_{22} = (TT^*)_{22}$, forces similarly

$$t_{2j} = 0, \quad j = 3, 4, \dots, n$$

continue

Continuing in this manner, we get that T is diagonal. ($T = D$).

(b) $\Rightarrow$ (a) Trivial because diagonal matrices are normal, and unitary similarities preserve normality.

(b) $\Rightarrow$ (c) Suppose $U^* A U = T$, which is diagonal as in Schur's Theorem,

so $T = \begin{bmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix}$

We recall a Theorem that says unitarily equivalent matrices have the same Frobenius norm:

$$\sum_{i,j=1}^{n} |a_{ij}|^2 = \sum_{i,j=1}^{n} |t_{ij}|^2$$

$$= \sum_{j=1}^{n} |\lambda_j|^2$$

(c) $\Rightarrow$ (b)

$$\sum_{i,j=1}^{n} |a_{ij}|^2 = \underbrace{\sum_{j=1}^{n} |\lambda_j|^2}_{\text{diag of } T} + \underbrace{\sum_{i<j} |t_{ij}|^2}_{\text{off-diag of } T}$$

when $U^* A U = T$ as per Schur.

$$\xRightarrow{(c)} \sum_{i<j} |t_{ij}|^2 = 0$$

$$\Rightarrow \quad t_{ij} = 0 \quad \text{for all } i<j$$

$\Rightarrow$ T is diagonal, i.e. (b) holds.

(b) $\Leftrightarrow$ (d) Easy!

$\downarrow$

Because the columns of unitary U such that $U^*AU = D$ are indeed orthonormal eigenvectors of A.